

Lesson 05.2: Overfitting

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References:

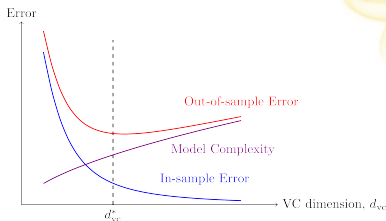
- Abu-Mostafa et al.: *“Learning From Data”* (2012)



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- ***Overfitting***
 - Bias Variance Analysis
 - Learning Curves

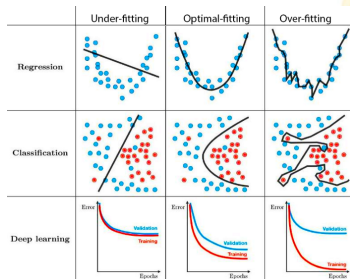
Overfitting: Definition

- **Overfitting** occurs when the model fits the data more than what is warranted
- Surpass point where E_{out} is minimal (optimal fit)
 - Model complexity too high for data/noise
 - Noise in training set mistaken for signal
- **Fitting noise instead of signal** is not useless but harmful
 - Model infers in-sample pattern that, when extrapolated out-of-sample, deviates from target function $\Rightarrow$ poor generalization



Optimal Fit

- The opposite of overfitting is **optimal fit**
 - Train a model with proper complexity for the data
- The optimal fit:
 - Implies E_{out} is minimal
 - Does not imply generalization error $E_{out} - E_{in}$ is minimal
 - E.g., no training implies generalization error equal to 0
- The **generalization error** is the additional error $E_{out} - E_{in}$ when going from in-sample to out-of-sample



Overfitting: Diamond Price Example

- Predict diamond price as a function of carat size (regression problem)
- True relationship**

$$\text{price} \sim (\text{carat size})^2 + \varepsilon$$

where:

- Square function: price increases more with rarity
- Noise ε : e.g., market noise, missing features
- Fit with:**

- Line*

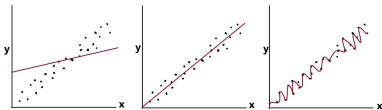
- Underfit
- High bias (large error)
- Low variance (stable model)

- Polynomial of degree 2*

- right fit

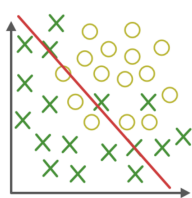
- Polynomial of degree 10*

- Overfit (wiggly curve)
- Low bias
- High variance (many degrees of freedom)

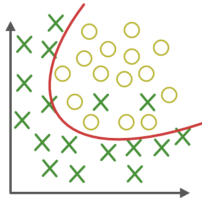


Overfitting: Classification Example

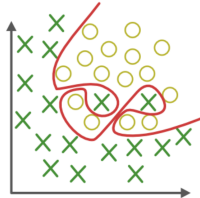
- Assume:
 - You want to separate 2 classes using 2 features x_1, x_2
 - The true class boundary has a parabola shape
- You can use logistic regression and a decision boundary equal to:
 - A line $\text{logit}(w_0 + w_1x + w_2y)$
 - Underfit
 - High bias, low variance
 - A parabola $\text{logit}(w_0 + w_1x + w_2x^2 + w_3xy + w_4y^2)$
 - Right fit
 - A wiggly decision boundary $\text{logit}(w_0 + \text{high powers of } x_1, x_2)$
 - Overfit
 - Low bias, high variance



Under-fitting



Appropriate-fitting



Over-fitting

- 
- Overfitting
 - ***Bias Variance Analysis***
 - Learning Curves

VC Analysis vs Bias-Variance Analysis

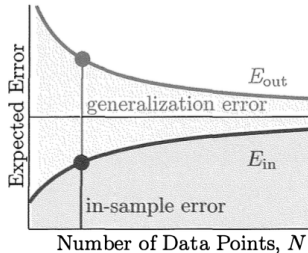
- Both VC analysis and bias-variance analysis are concerned with the hypothesis set $\mathcal{H}$

- VC analysis:**

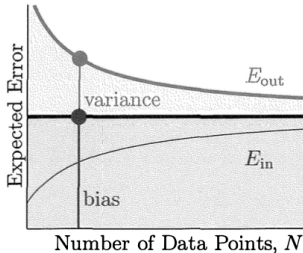
$$E_{out} \leq E_{in} + \Omega(\mathcal{H})$$

- Bias-variance analysis**

$$E_{out} = \text{bias} + \text{variance}$$



VC Analysis



Bias-Variance Analysis

Hypothesis Set and Bias-Variance Analysis

- **Learning** consists in finding $g \in \mathcal{H}$ such that $g \approx f$ where f is an unknown function
- The **tradeoff in learning** is between:
 - Bias vs variance
 - Overfitting vs underfitting
 - More complex vs less complex $\mathcal{H} / h$
 - Approximation (in-sample) vs generalization (out-of-sample)

Decomposing Error in Bias-Variance (1/4)

- **Problem**

- Regression set-up: target is a real-valued function
- Hypothesis set $\mathcal{H} = \{h_1(\underline{\mathbf{x}}), h_2(\underline{\mathbf{x}}), \dots, h_n(\underline{\mathbf{x}})\}$
- Training data $\mathcal{D}$ with N examples
- Squared error $E_{out} = \mathbb{E}[(g(\underline{\mathbf{x}}) - f(\underline{\mathbf{x}}))^2]$
- Choose the best function $g \in \mathcal{H}$ that approximates unknown f

- **Question**

- What is the out-of-sample error $E_{out}(g)$ as function of $\mathcal{H}$ for a training set of N examples?

Decomposing Error in Bias-Variance (2/4)

- The final hypothesis g depends on training set D , so make the dependency explicit $g^{(D)}$:

$$E_{out}(g^{(D)}) \triangleq \mathbb{E}_{\underline{\mathbf{x}}}[(g^{(D)}(\underline{\mathbf{x}}) - f(\underline{\mathbf{x}}))^2]$$

- Interested in:
 - Hypothesis set $\mathcal{H}$ rather than specific h
 - Training set D of N examples, rather than a specific D
- Remove dependency from D by averaging over all possible training sets D with N examples:

$$E_{out}(\mathcal{H}) \triangleq \mathbb{E}_D[E_{out}(g^{(D)})] = \mathbb{E}_D[\mathbb{E}_{\underline{\mathbf{x}}}[(g^{(D)}(\underline{\mathbf{x}}) - f(\underline{\mathbf{x}}))^2]]$$

Decomposing Error in Bias-Variance (3/4)

- Switch the order of the expectations since the quantity is non-negative:

$$E_{out}(\mathcal{H}) = \mathbb{E}_{\underline{x}}[\mathbb{E}_D[(g^{(D)}(\underline{x}) - f(\underline{x}))^2]]$$

- Focus on $\mathbb{E}_D[(g^{(D)}(\underline{x}) - f(\underline{x}))^2]$ which is a function of $\underline{x}$
- Define the *average hypothesis* over all training sets as:

$$\bar{g}(\underline{x}) \triangleq \mathbb{E}_D[g^{(D)}(\underline{x})]$$

- Add and subtract it inside the $\mathbb{E}_D$ expression:

$$\begin{aligned} E_{out}(\mathcal{H}) &= \mathbb{E}_{\underline{x}} \left[\mathbb{E}_D \left[(g^{(D)}(\underline{x}) - f(\underline{x}))^2 \right] \right] \\ &= \mathbb{E}_{\underline{x}} \mathbb{E}_D [(g^{(D)} - \bar{g} + \bar{g} - f)^2] \\ &= \mathbb{E}_{\underline{x}} \mathbb{E}_D [(g^{(D)} - \bar{g})^2 + (\bar{g} - f)^2 + 2(g^{(D)} - \bar{g})(\bar{g} - f)] \\ &\quad (\mathbb{E}_D \text{ is linear and } (\bar{g} - f) \text{ doesn't depend on } D) \\ &= \mathbb{E}_{\underline{x}} [\mathbb{E}_D [(g^{(D)} - \bar{g})^2] + (\bar{g} - f)^2 + 2\mathbb{E}_D [(g^{(D)} - \bar{g})](\bar{g} - f)] \end{aligned}$$

Decomposing Error in Bias-Variance (4/4)

- The cross term:

$$\mathbb{E}_D[(g^{(D)} - \bar{g})(\bar{g} - f)]$$

disappears since applying the expectation on D , it is equal to:

$$(g^{(D)} - \mathbb{E}_D[\bar{g}])(\bar{g} - f) = 0 \cdot (\bar{g} - f) = 0 \cdot \text{constant}$$

- Finally:

$$\begin{aligned} E_{out}(\mathcal{H}) &= \mathbb{E}_{\underline{x}}[\mathbb{E}_D[(g^{(D)} - \bar{g})^2] + (\bar{g}(\underline{x}) - f(\underline{x}))^2] \\ &= \mathbb{E}_{\underline{x}}[\mathbb{E}_D[(g^{(D)} - \bar{g})^2]] + \mathbb{E}_{\underline{x}}[(\bar{g} - f)^2] \quad (\mathbb{E}_{\underline{x}} \text{ is linear}) \\ &= \mathbb{E}_{\underline{x}}[\text{var}(\underline{x})] + \mathbb{E}_{\underline{x}}[\text{bias}(\underline{x})^2] \\ &= \text{variance} + \text{bias} \end{aligned}$$

Interpretation of Average Hypothesis

- The **average hypothesis** over all training sets

$$\bar{g}(\underline{x}) \triangleq \mathbb{E}_D[g^{(D)}(\underline{x})]$$

can be interpreted as the “best” hypothesis from $\mathcal{H}$ training on N samples

- Note: $\bar{g}$ is not necessarily $\in \mathcal{H}$
- In fact it's like **ensemble learning**:
 - Consider all the possible data sets D with N samples
 - Learn g from each D
 - Average all the hypotheses

Interpretation of Variance and Bias Terms

- The out-of-sample error can be decomposed as:

$$E_{out}(\mathcal{H}) = \text{bias}^2 + \text{variance}$$

- Bias term**

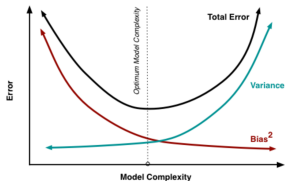
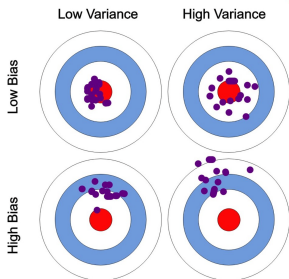
$$\text{bias}^2 = \mathbb{E}_{\underline{x}}[(\bar{g}(\underline{x}) - f(\underline{x}))^2]$$

- Does not depend on learning as it is not a function of the data set D
- Measures how limited $\mathcal{H}$ is
 - I.e., the ability of $\mathcal{H}$ to approximate the target with infinite training sets

- Variance term**

$$\text{variance} = \mathbb{E}_{\underline{x}} \mathbb{E}_D[(g^{(D)}(\underline{x}) - \bar{g}(\underline{x}))^2]$$

- Measures variability of the learned hypothesis from D for any $\underline{x}$
 - With infinite training sets, we could focus on the “best” g , which is $\bar{g}$
 - But we have only one data set D at a time, incurring a cost



Variance and Bias Term Varying Cardinality of $\mathcal{H}$

- If hypothesis set **has a single function**:

$$\mathcal{H} = \{h \neq f\}$$

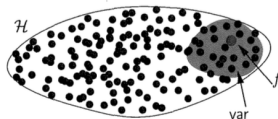
- Large bias
 - h might be far from f
- Variance = 0
 - No cost in choosing hypothesis



- If hypothesis set **has many functions**:

$$\mathcal{H} = \{\text{many hypotheses } h\}$$

- Bias can be 0
 - E.g., if $f \in \mathcal{H}$
- Large variance
 - Depending on data set D , end up far from f
 - Larger $\mathcal{H}$, farther g from f



Bias-Variance Trade-Off: Numerical Example

- **Machine learning problem:**
 - Target function $f(x) = \sin(\pi x)$, $x \in [-1, 1]$
 - Noiseless target
 - You have $f(\underline{x})$ for $N = 2$ points
- **Two hypotheses sets $\mathcal{H}$:**
 - Constant model: $\mathcal{H}_0 : h(x) = b$
 - Linear model: $\mathcal{H}_1 : h(x) = ax + b$
- **Which model is best?**
 - Depends on the perspective!
 - Best for *approximation*: minimal error approximating the sinusoid
 - Best for *learning*: learn the unknown function with minimal error from 2 points

Bias-Variance Trade-Off: Numerical Example

- Approximation**

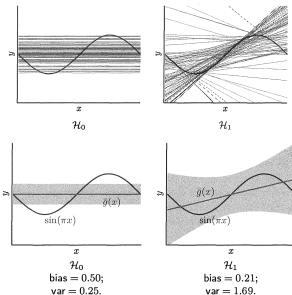
- $E_{out}(g_0) = 0.5$
 - g_0 is a constant and approximates the sinusoid poorly (higher bias)
 - $E_{out}(g_1) = 0.2$
 - g_1 is a line and has more degrees of freedom (lower bias)
 - The line model *approximates better* than the constant model

- Learning**

- Algorithm:
 - Pick 2 points as training set D
 - Learn g from D
 - Compute $\mathbb{E}_D[E_{out}(g)]$
 - Average over all data sets D :

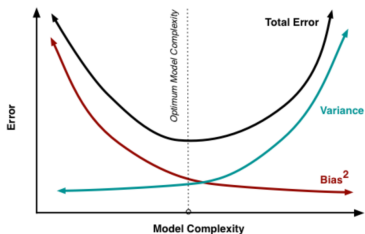
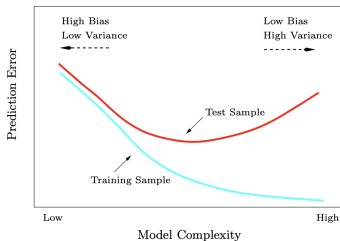
$$E_{out} = \text{bias}^2 + \text{variance}$$

- $E_{out}(g_0) = 0.5 + 0.25 = 0.75$
 - g_0 is more stable given the data set (lower variance)
 - $E_{out}(g_1) = 0.2 + 1.69 = 1.9$
 - g_1 heavily depends on the training set (higher variance)



Bias-Variance Curves

- **Bias-variance curves** are plots of E_{out} increasing the complexity of the model
- Typical **shape** of bias-variance curves
 - E_{in} and E_{out} start from the same point
 - E_{in}
 - Is decreasing with increasing model complexity
 - Can even go to 0
 - E_{out}
 - Is always larger than E_{in}
 - Is the sum of bias and variance
 - Has a bowl shape
 - Reaches a minimum for optimal fit
 - Before the minimum there is a “high bias / underfitting” regime
 - After the minimum there is a “high variance / overfitting” regime

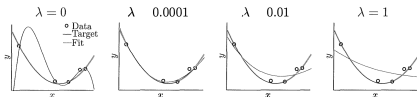


Bias-Variance Curves and Regularization

- Use **model with regularization** to learn at the same time:
 - Model coefficients $\underline{w}$
 - Model “complexity” (e.g., VC dimension)
- Learn the optimal model $\underline{w}(\lambda)$ as a function of $\lambda = \{..., 10^{-1}, 1.0, 10, ...\}$ by optimizing:

$$\underline{w}(\lambda) = \operatorname{argmin}_{\underline{w}} E_{aug}(\underline{w}) = E_{in}(\underline{w}) + \Omega(\lambda)$$

- Interpret regularization parameter:
 - Small λ :
 - Complex model
 - Low bias
 - High variance
 - Large λ :
 - Simple model
 - High bias
 - Low variance
 - An intermediate λ is optimal



How to Measure the Model Complexity

- Number of features
- Parameters for model form / degrees of freedom, e.g.,
 - VC dimension d_{VC}
 - Degree of polynomials
 - k in KNN
 - ν in NuSVM
- Regularization param λ
- Training epochs for neural network

Bias-Variance Decomposition with a Noisy Target

- Extend bias-variance decomposition to **a noisy target**

$$y = f(\underline{\mathbf{x}}; \underline{\mathbf{w}}) + \varepsilon = \underline{\mathbf{w}}^T \underline{\mathbf{x}} + \varepsilon$$

- With similar hypothesis and analysis conclude that:

$$\begin{aligned} E_{out}(\mathcal{H}) &= \mathbb{E}_{D, \underline{\mathbf{x}}} \left[(g^{(D)} - \bar{g})^2 \right] + \mathbb{E}_{\underline{\mathbf{x}}} \left[(\bar{g} - f)^2 \right] + \mathbb{E}_{\varepsilon, \underline{\mathbf{x}}} \left[(f - y)^2 \right] \\ &= \text{variance} + \text{bias} + \text{stochastic noise} \end{aligned}$$

- The out-of-sample error is the sum of 3 contributions
 1. **Variance**: from the set of hypotheses to the centroid of the hypothesis set
 2. **Bias**: from the centroid of the hypothesis set to the noiseless function
 3. **Noise**: from the noiseless function to the real function

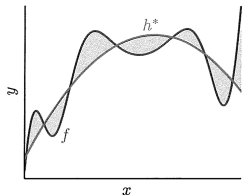
Bias as Deterministic Noise

- The bias term can be interpreted as **deterministic noise**
- **Bias** is the part of the target function that hypothesis set cannot capture:

$$h^*(\underline{x}) - f(\underline{x})$$

where:

- $h^*(\cdot)$ is the best approximation of $f(\underline{x})$ in the hypothesis set $\mathcal{H}$ (e.g., $\bar{g}(x)$)
- The hypothesis set $\mathcal{H}$ cannot learn the deterministic noise since it is outside of its ability, and thus it behaves like “noise”

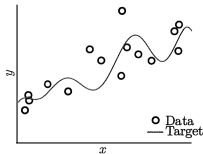


Deterministic vs Stochastic Noise in Practice

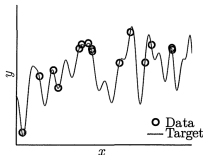
- **Deterministic noise:**
 - Fixed for a particular $\underline{x}$
 - Depends on $\mathcal{H}$
 - Independent of ε or D
- **Stochastic noise:**
 - Not fixed for $\underline{x}$
 - Independent of D or $\mathcal{H}$
- In a machine learning problem, no difference exists between stochastic and deterministic noise, as $\mathcal{H}$ and D are fixed
 - From the training set alone, you cannot determine if data is from a *noiseless complex* target or a *noisy simple* target

Deterministic vs Stochastic Noise Example

- **Two targets:**
 - Noisy low-order target (10th order polynomial)
 - Noiseless high-order target (50th order polynomial)
- **Training set**
 - Generate $N = 15$ data points
- **Two models**
 - $\mathcal{H}_2$ low-order hypothesis (2nd order polynomial)
 - $\mathcal{H}_{10}$ high-order hypothesis (10th order polynomial)
- Learning model:
 - Sees only training samples
 - Can't distinguish noise sources



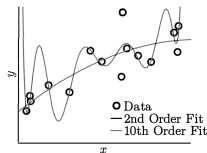
(a) 10th order target function



(b) 50th order target function

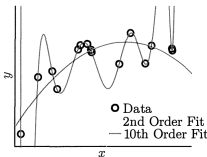
Deterministic vs Stochastic Noise Example

- Noisy low-order target:
 - Fit 2nd to 10th order polynomial
 - $\downarrow E_{in}$ (more degrees of freedom), $\uparrow\uparrow E_{out}$ (fits noise)
 - Noiseless high-order target
 - Fit 2nd to 10th order polynomial
 - Same phenomenon
 - $\downarrow E_{in}$ (more degrees of freedom), $\uparrow\uparrow E_{out}$ (fits noise)
 - Wrong approach**
 - Target is 10th order polynomial
 - Use 10-order hypothesis to fit target perfectly
 - Right approach**
 - Consider number of data points, i.e., 15 points
 - Rule of thumb from VC analysis
- degrees of freedom of model = number of data point
- Use 1 or 2 degrees of freedom



(a) Noisy low order target

10th order noisy target		
	2nd Order	10th Order
E_{in}	0.050	0.034
E_{out}	0.127	9.00



(b) Noiseless high order target

50th order noiseless target		
	2nd Order	10th Order
E_{in}	0.029	10^{-5}
E_{out}	0.120	7680

Amount of Data and Model Complexity

- One must match the *model complexity* to
 - **Data resources**
 - **Signal to noise ratio**
 - **Not target complexity**
- The rule of thumb is:

$$d_{VC}(\text{degrees of freedom of the model}) = N(\text{number of data points})/10$$

- In other words, 10 data points needed to fit a degree of freedom
- If the data is noisy, you need even more data

Overfitting as a Function of Data Resources, Model Complexity, Noise

- You can measure overfitting in terms of **generalization error** $\frac{E_{out} - E_{in}}{E_{out}}$
 - $\uparrow$ data resources $N \implies \downarrow$ overfitting
 - $\uparrow$ model complexity $d_{VC} \implies \uparrow$ overfitting
 - $\uparrow$ deterministic noise (target complexity) $\implies \uparrow$ overfitting
 - $\uparrow$ stochastic noise $\sigma^2 \implies \uparrow$ overfitting

- 
- Overfitting
 - Bias Variance Analysis
 - *Learning Curves*

Learning Curves vs Bias-Variance Curves

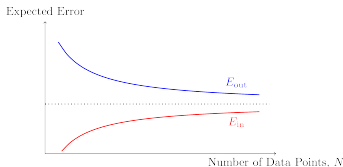
- **Learning curves** are the dual of the bias-variance curves
- For **bias-variance curves**

$$E_{in}, E_{out} = f(d_{VC}|N)$$

- Keep the complexity of training set fixed (number of examples N)
- Vary the model in terms of:
 - Model complexity d
 - Number of features p
 - Regularization amount λ
- For **learning curves**

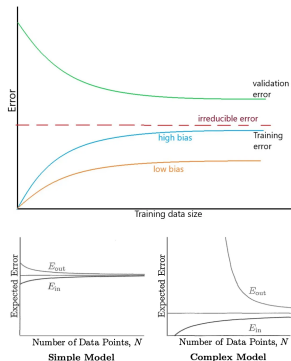
$$E_{in}, E_{out} = f(N|d_{VC})$$

- Fix the model
- Vary the size N of training set



Typical Form of Learning Curves

- Learning curves plot E_{in} and E_{out} as a function of data amount N , given the model $h()$
 - $E_{out} \geq E_{in}$ for any N
- Small N**
 - E_{in} is small (even 0)
 - The model is likely overfitted, memorizing and generalizing poorly
 - E_{out} is large
- Increasing N**
 - $E_{in} \uparrow$ as the model cannot fit all data
 - $E_{out} \downarrow$ as the model fits better and generalizes better
 - Generalization error $E_{out} - E_{in} \downarrow$
- $N \rightarrow \infty$
 - E_{in} reaches a maximum (irreducible error)
 - E_{out} reaches a minimum
 - $E_{out} - E_{in}$ depends on the complexity of the model



High-Bias vs High-Variance Regime

- From the learning curve you can see the two regimes:
 - High-variance regime** for small N
 - E_{in} is small
 - Small data set $D \implies$ high dependency on D
 - More data helps reduce gap between E_{in} and E_{out}
 - High-bias regime** for large N
 - E_{in} is large; flattens for large N
 - Best model fitted; more data won't help
 - E_{out} can be close to E_{in} (good generalization) or not

